**2. Occurrence**
Maximum synthesis occurs during climacteric ripening of fruits (*see* Box info) and tissues undergoing senescence. It is formed in almost all plant parts like roots, leaves, flowers, fruits and seeds.

**3. Transport in plants**
Ethylene can easily diffuse inside the plant through intercellular spaces.

**4. Precursor**
It is a derivative of amino acid methionine, linolenic acid and fumaric acid.

**5. Bioassay (Gas Chromatography)**
Ethylene can be measured by gas chromatography. This technique helps in the detection of exact amount of ethylene from different plant tissues like lemon and orange.

**6. Physiological Effects**
- Ethylene stimulates respiration and ripening in fruits.
- It breaks the dormancy of buds, seeds and storage organs.
- It stimulates formation of abscission zone in leaves, flowers and fruits. This makes the leaves to shed prematurely.
- Inhibition of stem elongation (shortening the internode).
- Growth of lateral roots and root hairs. This increases the absorption surface of the plant roots.
- Ethylene normally reduces flowering in plants except in Pine apple and Mango.

### 15.2.5 Abscisic Acid (ABA) (Stress Phyto hormone)

**1. Discovery**
In 1963, the hormone was first isolated by **Addicott *et al.,*** from young cotton bolls and named as **Abscission II**. Eagles and Wareing during 1963–64 isolated a dormancy inducing substance from leaves of *Betula* and called it as dormin. In 1965, it was found by Cornsforth *et al.*, that both dormin and abscission are chemically same compounds and called **Abscisic Acid (ABA)**.

**2. Occurrence**
This hormone is found abundantly inside the chloroplast of green cells.

**3. Precursors**
The hormone is formed from mevalonic acid pathway or xanthophylls.

**4. Transport in plants**
Abscisic acid is transported to all parts of the plant through diffusion as well as through phloem and xylem.

**5. Chemical structure**
It has carotenoid structure.

**6. Bioassay (Rice Coleoptile)**
The inhibition of IAA induces straight growth of rice seedling coleoptiles.

**7. Physiological effects**
- It helps in reducing transpiration rate by closing stomata.
- ABA is a powerful growth inhibitor. It causes 50% inhibition of growth in Oat coleoptile.
- It induces bud and seed dormancy.
- It promotes the abscission of leaves, flowers and fruits by forming abscission layers.
- ABA plays an important role in plants during water stress and during drought conditions. It results in loss of turgor and closure of stomata.
- In Cannabis sativa, induces male flower formation on female plants.
- It promotes sprouting in storage organs like Potato.
- It inhibits the shoot growth and promotes growth of root system. This character protect the plants from water stress. Hence, ABA is called as **stress hormone.**

## 15.3 Plant Movements




Plants have the capacity for changing their positions in response to external or internal stimuli, which are known as **plant movements**. Movements are basically of two types: I. Vital movements and II. Physical movements (hygroscopic) (Figure 15.10).

### I. Vital movements

Vital movements are those which are exhibited by the living cells or plants or



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